

# SHIVAM'S WORLD

Interactive Game-Based Portfolio

## TECHNICAL REQUIREMENTS DOCUMENT

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|----------|---------------------------|
| Version: | 1.0                       |
| Owner:   | Shivam Singh              |
| Date:    | June 14, 2026             |
| Status:  | Approved / Technical Spec |

## 1. Project Overview

- **Project Name:** Shivam's World
- **Type:** Interactive Game-Based Portfolio
- **Platform:** Web Application
- **Architecture:** Single Page Application (SPA)
- **Navigation:** Scene-based, click-driven (no traditional scrolling)

## 2. Objectives

- Create a premium, game-like portfolio menu system to drive visitor interaction.
- Showcase Shivam's technical projects, skills, timeline, achievements, and contact portals.
- Ensure immersive audio feedback, hover effects, and organic transition animations.

## 3. Technical Stack

- **Frontend:** Next.js (App Router), React, TypeScript
- **Styling:** Tailwind CSS, custom CSS Variables (flat colors, no gradients)
- **Animation:** Framer Motion (scene swaps), GSAP (zoom timelines, entry portals)
- **Audio:** Howler.js (cache/preload sounds, local volume states)
- **Deployment & Git:** Vercel deployments synced with Git + GitHub

## 4. Application Architecture & Scene Flow

Loading Screen (Ambience, Progress Bar, Load Completion Chime)

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Profile Unlock Screen (Player Found Portrait, Scale Hover, Click to Unlock)

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World Entry Scene (Portal transition zoom effect, durations: 2-3s)

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Main Hub Central Control (Origin Story, Quest Log, Ability Tree, Adventure Path, Portal)

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Achievements Sub-system (Accessed via Journey Adventure Path)

## 5. Scene Specifications

### 5.1. Loading Screen

- Preloads assets over a fixed 3-second progress duration.
- Displays the branding logo, progress bar (0% -> 100%), and dynamic text.
- Audio: Loops loading ambience and triggers a chime upon completion.

### 5.2. Profile Unlock Screen

- Features portrait graphics and a text overlay reading 'Player Found'.
- Click to Continue button triggers zoom animation, unlock sound, and loads world entry.

### 5.3. World Entry Screen

- Transition zoom overlay reading 'Entering Shivam's World'.
- Duration: 2-3 seconds, with immersive portal sound effects.

## 5.4. Main Hub Screen

- Central dashboard featuring 5 interactive navigation menu cards:
  1. Origin Story (About Scene)
  2. Quest Log (Projects Scene)
  3. Ability Tree (Skills Scene)
  4. Adventure Path (Journey Scene)
  5. Communication Portal (Contact Scene)
- Hover: Scale 1.03 with glow border outline. Click: triggers scene-load and click ticks.

## 6. Projects System (Quest Log)

- Main Quest: CollabKaro (Featured Project card displaying: Title, Description, Role, Core Features, Tech Stack list, Demo link, and GitHub repository link).
- Side Quests: A grid of minor project cards supporting similar metadata fields.
- Audio triggers: Cinematic swoosh sound is triggered upon quest card selection.

## 7. Skills System (Ability Tree)

- Categories: Java, Web Development, React, DSA, AI, and Entrepreneurship.
- Presentation: Implemented as an Ability Tree mapping nodes, featuring expandable skill containers that expand via Framer Motion slide reveals.
- Audio: Knowledge unlock chime triggers when cards are toggled.

## 8. Journey (Adventure Path) & Achievements

- Milestone path tracking Levels 1 through 8 (Programming, HTML/CSS, JS, College, Projects, CollabKaro, AI, and Future Loading).
- Clicking checkpoints displays detail modal overlays with checkpoint sounds.
- View Achievements CTA links directly to Achievements system, revealing achievements: Founder of CollabKaro, Built Real Projects, Startup Builder, and Continuous Learner.
- Achievements are visual cards which play dedicated unlock fanfares on select.

## 10. Contact System (Communication Portal)

- Final Quest title with responsive inputs: Name, Email, and message textbox.
- Social coordinates for GitHub, LinkedIn, Instagram, and Email.
- CTA display: 'Let's Build Something Amazing Together'. Audio: Loops portal ambience.

## 11. Audio Design Specs

- Global preferences: Sound Toggle (On/Off) and log Volume Slider stored in Local Storage.
- Core sound channels managed by Howler.js context:
  - \* Ambient: loader loop, main hub soundtrack, contact portal loop.
  - \* Interactive: UI hover, click down ticks, mission load, checkpoint chimes, triumphs.

## 12. Design System Tokens

| Token Name     | Hex Value | Description         |
|----------------|-----------|---------------------|
| Background     | #050505   | Canvas background   |
| Surface        | #101010   | Panel backgrounds   |
| Cards          | #151515   | Card structures     |
| Borders        | #242424   | Separator borders   |
| Text           | #FFFFFF   | Primary title copy  |
| Secondary Text | #A3A3A3   | Secondary body text |

## 13. Typography & 14. Performance

- Typography: Inter (Primary) and Geist (Alternative). Minimalist and cinematic weight rules.
- Performance: Lighthouse score  $\geq 90$ . First content load  $< 3$  seconds.
- Image Optimization: Strictly WebP/AVIF file endings, dynamic lazy-loads for sub-scenes.

## 15. Future Roadmap

- Version 2: Hidden Developer Room (Konami/Terminal inputs), easter eggs, action achievements.
- Version 3: React Three Fiber interactive solar map, avatar control, multiplayer counters.

## 16. Success Criteria

- Visitors navigate and explore multiple scenes, rather than bouncing.
- Interactive audio, click feedbacks, and GSAP timelines make it read like a game interface.
- Extended visitor dwell time compared to classic vertical scroll portfolio formats.
- Effective technical delivery of Shivam's project details (CollabKaro) and coding history.